AURA Twin

History + wearable state.
Fresh-patient evidence.
A challengeable prediction.

AURA-SYNTHETIC-3.0.0

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Generated data only · No clinical validation or treatment recommendation

Give changing signals their historical context.

Current model aura-synthetic-3.0.0 · synthetic benchmark evidence

01

Historical context

Age, laboratory history and condition duration explain the baseline.

02

Changing observations

Glucose, HR, HRV, activity and sleep show recent state.

03

A reviewable forecast

One patient state supports prediction, context and evidence.

A clinician-oriented proof of concept; workflow benefit has not been measured.

Predict a defined event, at a defined time.

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EXACT-HORIZON EVENT

Glucose ≥180 mg/dL at t + 120 minutes

Not any excursion during the next two hours.

1 **Four forecasts**

+30 / +60 / +90 / +120 minutes

2 **One event probability**

Calibrated estimate for the exact two-hour reading

3 **Explicit convention**

70–180 is common CGM context; inclusive 180 is our label.

Context: NIDDK, Clinical Targets for Continuous Glucose Monitoring Data. Individual targets can differ.

Synthetic data with deliberately varied routines.

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01

Own seeded simulator

Wholly generated people and signals. No real records or device accounts.

02

Six families + one test

Six development routines; a separate withheld combination of familiar factors.

03

No hidden meal oracle

Future meals and simulator latent state are excluded from inputs.

Six development families; one withheld combination; new final identities.

Only the available past enters the prediction.

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01

Recent dynamics

Lagged CGM, slopes, variability, HR/HRV, steps and sleep.

02

Previous-day observations

Horizon-aligned readings are at t+h–24h.
Observed past, not today’s future.

03

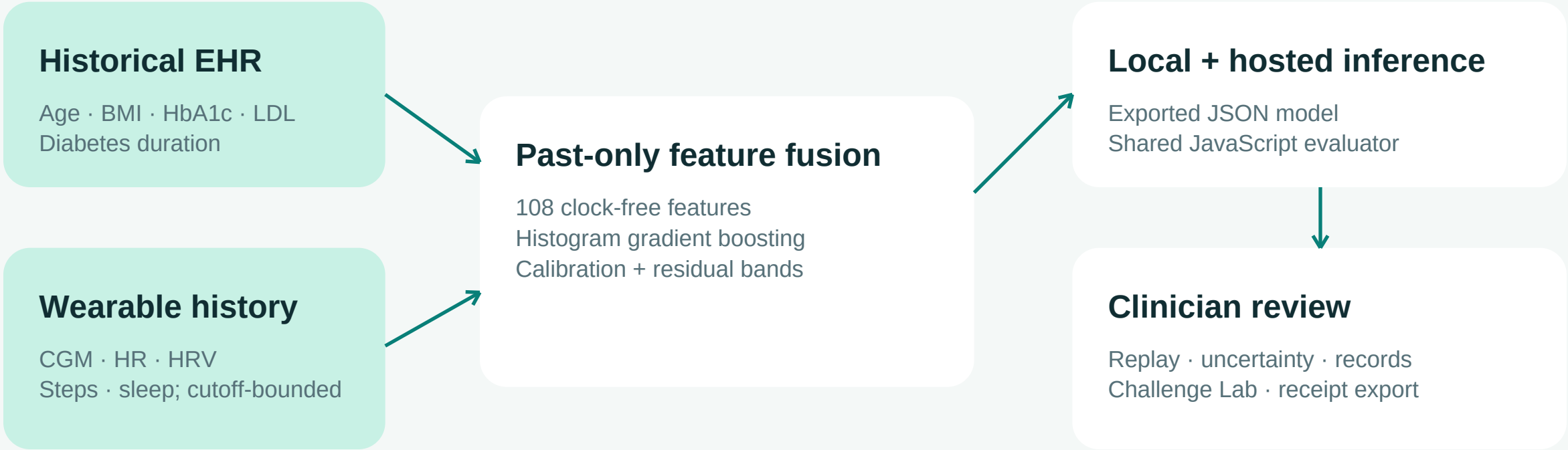
Explicit boundary

Changing future readings must not change a frozen prediction.

Past-only does not mean future behavior becomes predictable.

Two streams. One inspectable inference path.

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No future meals, future readings or hidden simulator state enter inference.

Freeze the policy before generating fresh final patients.

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768

Fit

Training identities
Candidate fitting only

80

Select

Separate identities
Choose candidate + policy shape

96

Calibrate

Separate identities
Probability, bands, threshold

120

Final test

New indices 2000–2119
Generate after model freeze

Every routine variant and window for an identity stays in one partition.

Six familiar families + one withheld factor combination; the same 120 final identities.

Select for the actual missed-versus-false tradeoff.

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01

Richer past observations

108 clock-free features; no actual future meals.

02

Task-aligned selection

Compare worst FP/FN ratios over overall and irregular groups.

03

One frozen policy

A single threshold set before fresh final evaluation.

Selection, calibration and final reporting have different patient partitions.

Measure error and the review-policy tradeoff.

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Frozen policy	Threshold	Recall	Precision	Missed +	False +
Frozen v2	0.1600	73.36%	35.46%	1,742	8,730
Current v3	0.1550	74.47%	37.34%	1,669	8,170

Primary counts improve; missed-window CI crosses zero

V3 +120 MAE 10.85 mg/dL · AUROC 0.934 · Brier 0.0400

Primary familiar-family test · overlapping windows, not unique clinical episodes.

Compare fixed policies on identical new windows.

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Routine	FN: v2 → v3	FP: v2 → v3	V3 recall	V3 precision
Regular	211 → 212	1,207 → 1,134	82.57%	46.96%
Meals +2h	164 → 159	1,366 → 1,370	85.64%	40.90%
Meals +4h	132 → 142	1,332 → 1,195	84.58%	39.46%
Jittered meals	491 → 452	1,541 → 1,384	65.26%	38.02%
Daily changes + skips	324 → 313	1,627 → 1,490	69.13%	31.99%
Variable sleep/activity	420 → 391	1,657 → 1,597	60.06%	26.91%

Withheld combination: missed positives 324 → 330 · false positives 1,657 → 1,624

Withheld combination is separate from this six-family primary table.

Show uncertainty without promising coverage.

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90.60%

Observed coverage

Nominal 90% descriptive residual bands.
Measured on the fresh final cohort.

61.8 mg/dL

Mean band width

At the two-hour horizon.
Broader uncertainty can accompany irregular behavior.

0.1550

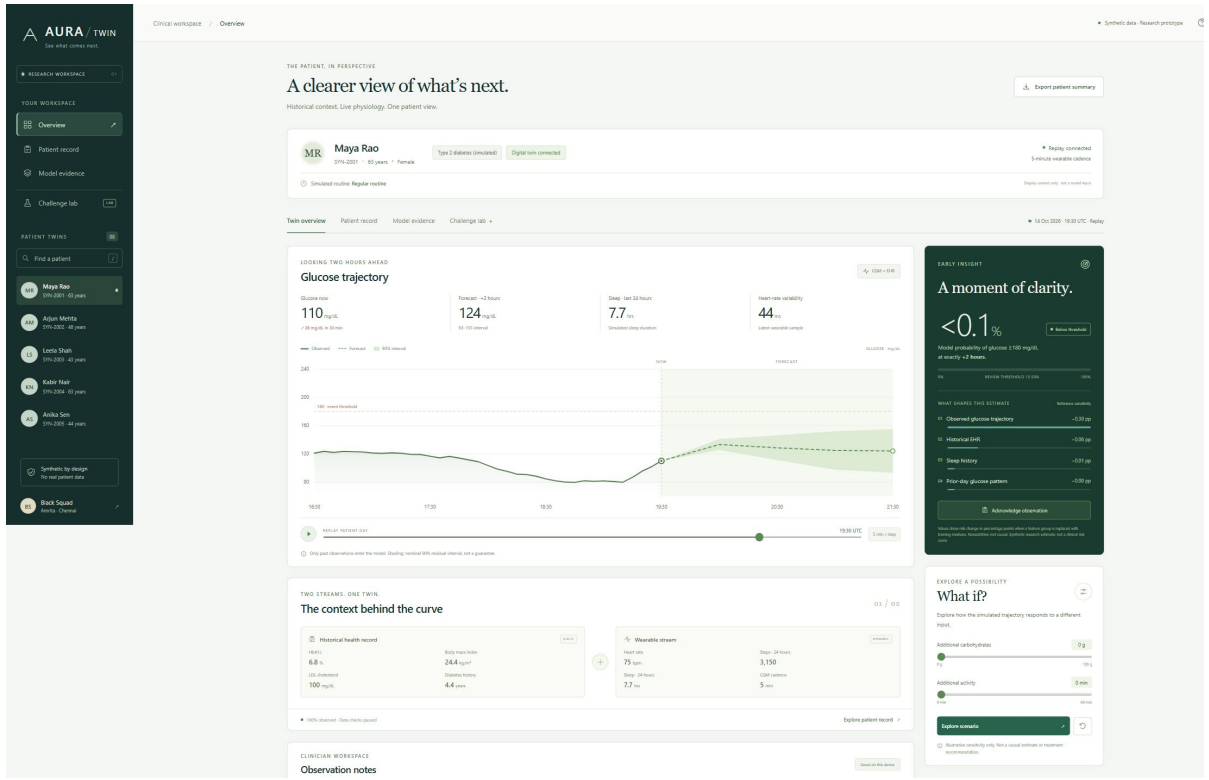
Review policy

Threshold fixed on calibration patients.
Not a validated clinical trigger.

Coverage is marginal. No guarantee per patient, subgroup or shifted population.

An overview designed for a clinical review.

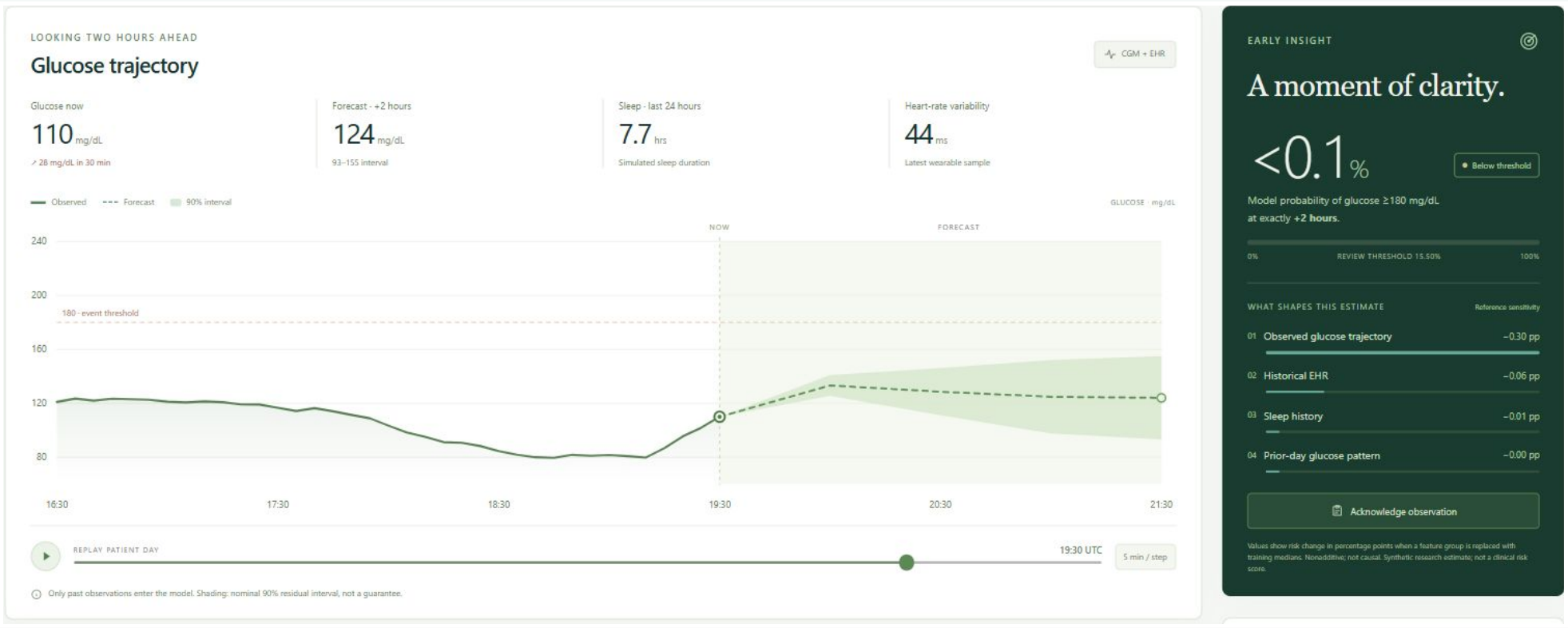
Current model aura-synthetic-3.0.0 · synthetic benchmark evidence



Actual local interface · all observations are generated for this demonstration

Replay makes the information boundary tangible.

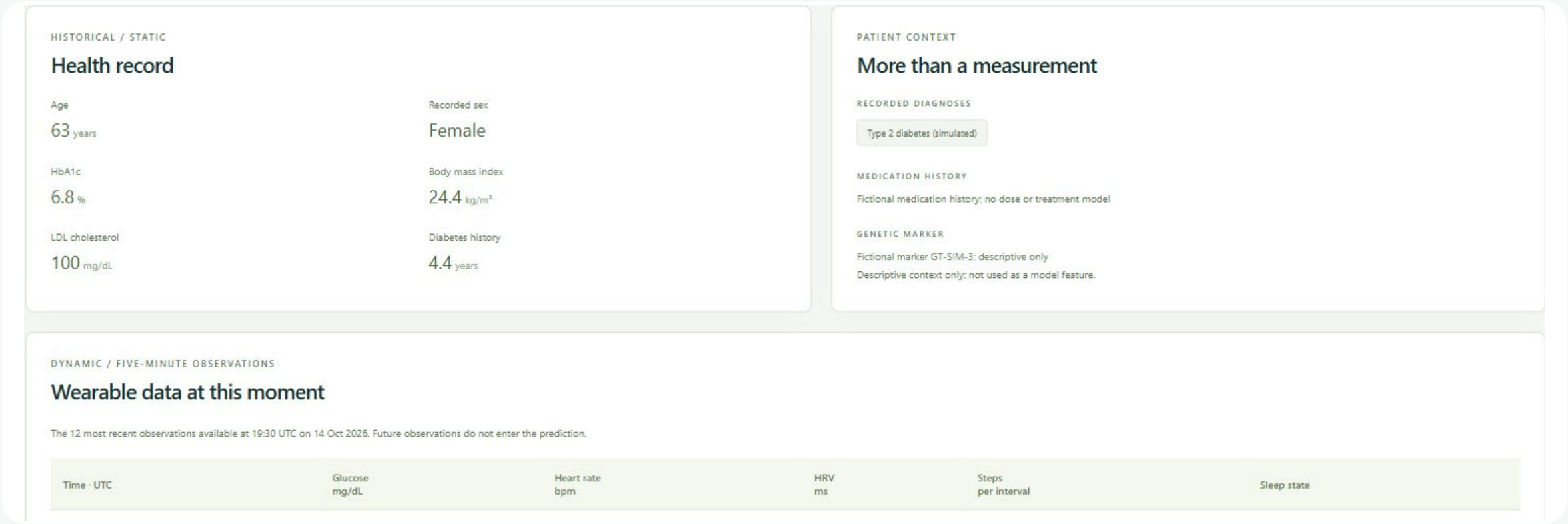
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Actual local interface · all observations are generated for this demonstration

Inspect context without turning it into a cause.

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Actual local interface · all observations are generated for this demonstration

Explore an assumption. Do not infer a treatment.

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01

Baseline forecast

The fitted model provides the numerical trajectory and event probability.

Reset restores this baseline.

02

Illustrative sensitivity

Meal/activity sliders apply hand-authored arithmetic assumptions.

Trained risk remains unchanged.

No future meal knowledge. No causal benefit estimate. No treatment or dosing recommendation.

Freeze the prediction. Then reveal the outcome.

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01

Choose

Patient, cutoff, sensor case.
Copy and truncate history.

02

Freeze

Predict before seeing outcomes.
Preserve output + hashes.

03

Reveal

Original recorded future.
No recomputation.

04

Compare

Model vs persistence.
Interval hits + receipt.

Public data: this enforces a computation boundary, not secure study blinding.

Four sensor cases expose different failure modes.

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Clean replay

An honest baseline; a local window may lose to persistence.

10-minute CGM gap

Forward fill permitted; show degraded quality.

30-minute dropout

Stale signal; withhold forecast and probability.

Undetected sensor bias

Plausible +30 mg/dL input shift may pass freshness checks.

The reference future stays unchanged. Persistence sees the same perturbed current value.

Make the experiment reproducible offline.

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01

What can be checked

Raw model bytes → SHA-256
Canonical input prefix → SHA-256
Recomputed forecast and revealed score
Mismatch detection for altered artifacts

02

What is not established

No digital signature
No independent timestamp
No trusted authorship guarantee
No clinical audit certification

```
node scripts/verify_receipt.mjs path/to/receipt.json
```

The evidence travels with the implementation.

Current model aura-synthetic-3.0.0 · synthetic benchmark evidence

01

Reproduce the model

Generator, selection protocol, calibration and frozen artifacts.

02

Verify implementation

Python/JavaScript parity, prefix invariance and challenge tests.

03

Run locally or hosted

One model evaluator; synthetic-only public demonstration.

Metrics and fingerprints identify the exact evaluated artifact.

A better benchmark is still a bounded claim.

Current model aura-synthetic-3.0.0 · synthetic benchmark evidence

01

Unobserved future behavior

Unexpected meals and abrupt behavior remain difficult to anticipate.

02

Synthetic assumptions

Designed physiology and routine families may favor the model.

03

Real-world evidence gap

No clinical, device, outcome or compliance validation.

Synthetic benchmark improvement is not clinical robustness.

Use the remaining failures to guide validation.

Current model aura-synthetic-3.0.0 · synthetic benchmark evidence

01

Broader tests

Independent simulators, held-out routine families and device faults.

02

Independent evaluation

Permitted external data, calibration, subgroups and silent studies.

03

Governed workflows

Clinical ownership, access control, audit and usability validation.

These are future validation steps, not implemented clinical capabilities.

A forecast you can inspect.

A working fusion pipeline.
A measured synthetic comparison.
A reproducible challenge.

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